Hassan Raza

AI Engineer | ML & Deep Learning | RAG & LLM Fundamentals | Python & Data Science
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SUMMARY

BS Artificial Intelligence student at KFUEIT with internship experience in AI-MERN Stack and Web development. Familiar with Python, Machine Learning fundamentals, data analysis, REST APIs, and web development technologies. Eager to apply technical skills, learn from industry professionals, and contribute to real-world projects.

TECHNICAL SKILLS

Machine Learning	Logistic Regression, Classification, Statistical Analysis, Supervised Learning, Model Evaluation
Data Science Libraries	Python, Pandas, NumPy, Scikit-Learn, Matplotlib, Exploratory Data Analysis (EDA)
RAG & LLM	LangChain, LlamaIndex (concepts), RAG Architectures, Chunking & Embedding Strategies,
LLM & AI APIs	GEMINI API, Hugging Face Transformers, Prompt Engineering, LLM
Python Development	Python Programming, OOP Concepts, Data Structures, Functions, Loops, Problem Solving, Basic Flask, REST API Fundamentals
Computer Vision & NLP	OpenCV, MediaPipe, NLP, Named Entity Recognition, Summarization, Speech Processing
DevOps & Tools	Git/GitHub, Docker(Basic), VS Code, Jupyter Notebook, Version Control
Frontend Integration	React.js, TypeScript, Tailwind CSS, Data Visualization Dashboards

PROFESSIONAL EXPERIENCE

AI / ML Engineer Intern — Hex Softwares Remote
Apr 2026 – May 2026 | 1.5 Months

- Applied core Machine Learning concepts including classification models, statistical analysis, and evaluation metrics to analyze and improve model performance on small-scale datasets.
- Worked with Python libraries such as Pandas and NumPy for data preprocessing, cleaning, and feature engineering to prepare datasets for machine learning experiments.
- Explored foundational concepts of Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to understand modern AI system design at an introductory level.
- Demonstrated a strong learning mindset by independently experimenting with ML/DL workflows.

AI-Integrated Full-Stack Intern — MercurySols Rahim Yar Khan, Pakistan
June 2025 – Sep 2025 | 3 Months

- Developed responsive and interactive frontend web applications using React.js, focusing on reusable components, UI design, and user experience improvements.
- Integrated RESTful APIs into frontend applications to fetch and display dynamic data from backend services in MERN projects.
- Implemented frontend features such as forms, dashboards, and dynamic pages to improve usability and real-world application structure.
- Explored and added simple AI-assisted functionalities in web applications to enhance user interaction and project innovation.

PROJECTS

Context-Aware AI Chatbot Engine (LangChain & Gemini API)

- Architected a RAG chatbot using LangChain and Gemini API to stream context-grounded responses on custom PDFs.
- Implemented optimized text chunking and Hugging Face embedding pipelines to maximize semantic search precision.
- Engineered strict system prompt structures and guardrails to eliminate hallucinations and maintain accuracy.
- Built a Python execution layer to automate backend system tasks directly via natural language queries.

Gesture-Controlled Chess Engine (OpenCV + MediaPipe)

- Developed an interactive computer vision application using Python to play chess entirely via real-time hand gestures.
- Integrated MediaPipe to track 21 hand landmarks for precise finger-tip positioning and gesture-to-action mapping.
- Utilized OpenCV to overlay dynamic bounding boxes, state messages, and control matrices onto the video feed.
- Optimized frame inference pipelines to maintain low-latency tracking and smooth interaction under varying lighting constraints.

NLP Queries Real Time Visualizer(React.js + FastAPI)

- Engineered a full-stack NLP playground to parse, analyze, and map syntactic structures in real time.
- Built high-performance asynchronous REST APIs using Python (FastAPI) for modular text pipeline execution.
- Implemented PCFG parsing algorithms to dynamically compute tree weights and resolve structural ambiguity.
- Integrated NLTK and spaCy modules to dynamically extract VerbNet thematic roles and WordNet clusters.

EDUCATION

Khwaja Fareed University of Engineering & IT (KFUEIT) Sep 2023 – Jun 2027
Bachelor of Science — Artificial Intelligence CGPA: 3.32 / 4.00
Rahim Yar Khan, Pakistan

- Currently in Final Year.
- Core coursework: Machine Learning, NLP, Statistical Analysis, Speech Processing, EDA, Parallel Computing.
- Strong hands-on foundation in Logistic Regression, classification models, and statistical analysis using Python, Pandas, NumPy, and Scikit-Learn.

CERTIFICATIONS

- Modern & Generative AI — Cisco Networking Academy
- Python Essentials — Cisco Networking Academy
- AI-MERN Stack Engineering — MercurySols (Internship Certification)

INTERESTS

- Chess (Problem solving, Creative thinking)
- Video Editing (Content Creation)
- Analyzing algorithmic strategy mechanics